Threat Intelligence Integration

Introduction

The objective of this practical task is to integrate external threat intelligence feeds into a Security Information and Event Management (SIEM) environment and demonstrate how threat intelligence can enhance security monitoring. This exercise involves integrating a threat intelligence feed from **AlienVault Open Threat Exchange** into **Wazuh**, generating a test alert using a mock IP address, enriching the alert with threat intelligence data, and performing threat hunting to detect suspicious activities within the system logs.

Tools Used

Tool	Purpose
Wazuh	Security monitoring and log analysis platform
AlienVault OTX	Threat intelligence feed provider
TheHive	Incident response and case management platform

Threat Feed Import

The first step involved importing threat intelligence from **AlienVault Open Threat Exchange** into the Wazuh SIEM environment.

An account was created on the OTX platform and an API key was generated to allow Wazuh to access the threat intelligence feed. The OTX integration script (`custom-otx.py`) was configured inside the Wazuh integrations directory.

The script enables Wazuh to query OTX for indicators of compromise (IOCs) such as malicious IP addresses and enrich alerts with threat intelligence data.

The integration script was verified using the following command:

```
sudo ls /var/ossec/integrations
```

The output confirmed that the integration script was successfully installed.

Generate a Test IOC Event

To test the integration, a simulated malicious event containing the IP address **192.168.1.100** was generated. This was done by inserting a failed SSH login attempt into the system authentication logs. The command injected a log entry into `/var/log/auth.log`, which is monitored by Wazuh.

Example simulated log event:

```
Failed password for invalid user admin from 192.168.1.100 port
22 ssh2
```

After the log was inserted, Wazuh detected the event and processed it as a security alert.

Alert Enrichment with Threat Intelligence

Once the alert was detected, the IP address **192.168.1.100** was analyzed using the **AlienVault Open Threat Exchange** dashboard.

The IP was searched within the OTX platform to verify its reputation and determine whether it is associated with malicious activity. OTX provides information such as:

- Known malicious indicators
- Associated threat pulses
- Reputation data
- Related threat campaigns

The threat intelligence lookup demonstrated how external intelligence feeds can provide additional context for detected alerts.

Alert Documentation

The enriched alert was documented as shown below:

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Linked to C2 server

This table documents the indicator of compromise detected during the experiment and the reputation information obtained from OTX.

Threat Hunting

Threat hunting was performed in the Wazuh dashboard to detect suspicious account activity related to the **Valid Accounts** technique from the **MITRE ATT&CK** framework.

The following query was used in the Discover search bar:

```
NOT user.name:"system"
```

This query identifies login activities performed by user accounts other than system accounts. Such queries are commonly used by SOC analysts to detect suspicious login attempts, compromised credentials, or unauthorized account activity.

The results showed authentication-related events in the logs, demonstrating how threat hunting can be used to proactively identify potential security incidents.

OTX Dashboard Verification

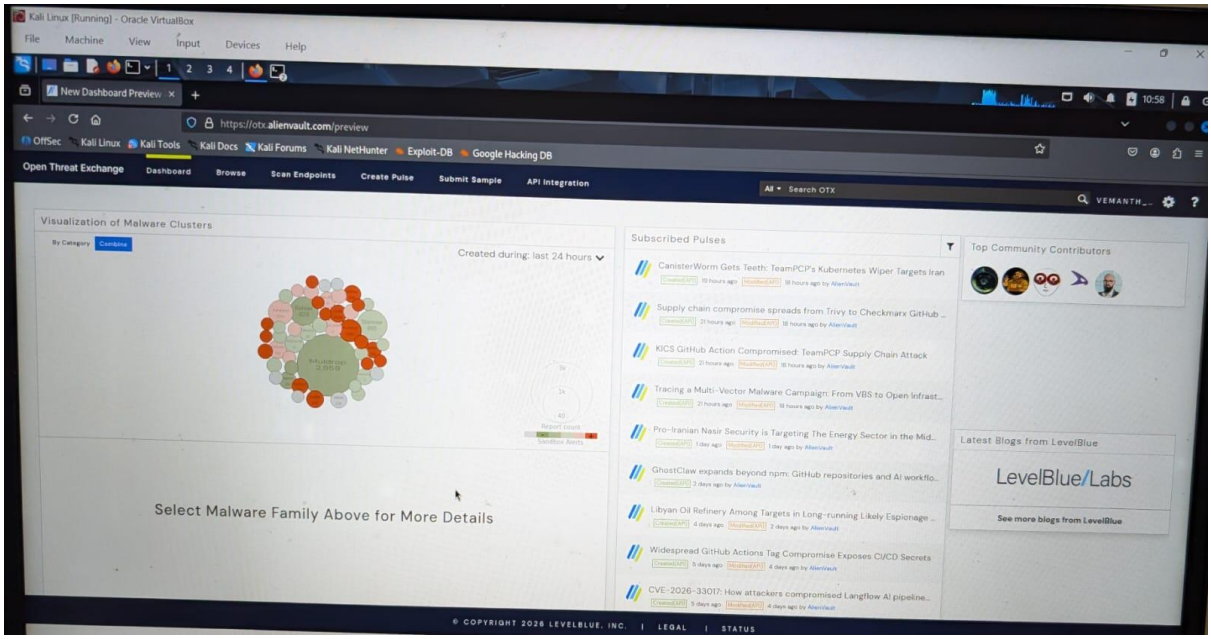
The OTX dashboard was accessed to verify the threat intelligence environment and confirm successful integration with the SIEM platform.

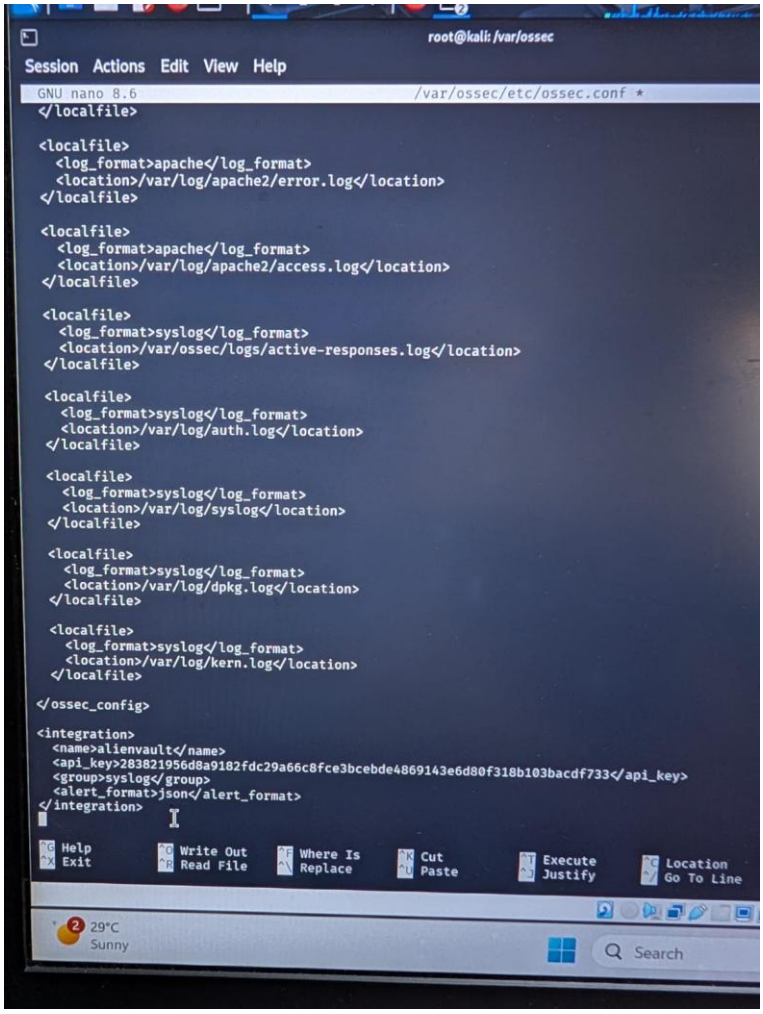
Findings

Threat intelligence integration enabled Wazuh to correlate security alerts with external indicators from AlienVault OTX. A simulated SSH login attempt from the IP address 192.168.1.100 generated a security alert. The alert was enriched with threat intelligence information, demonstrating how SOC analysts can improve threat detection, investigation, and response using external intelligence feeds.

Conclusion

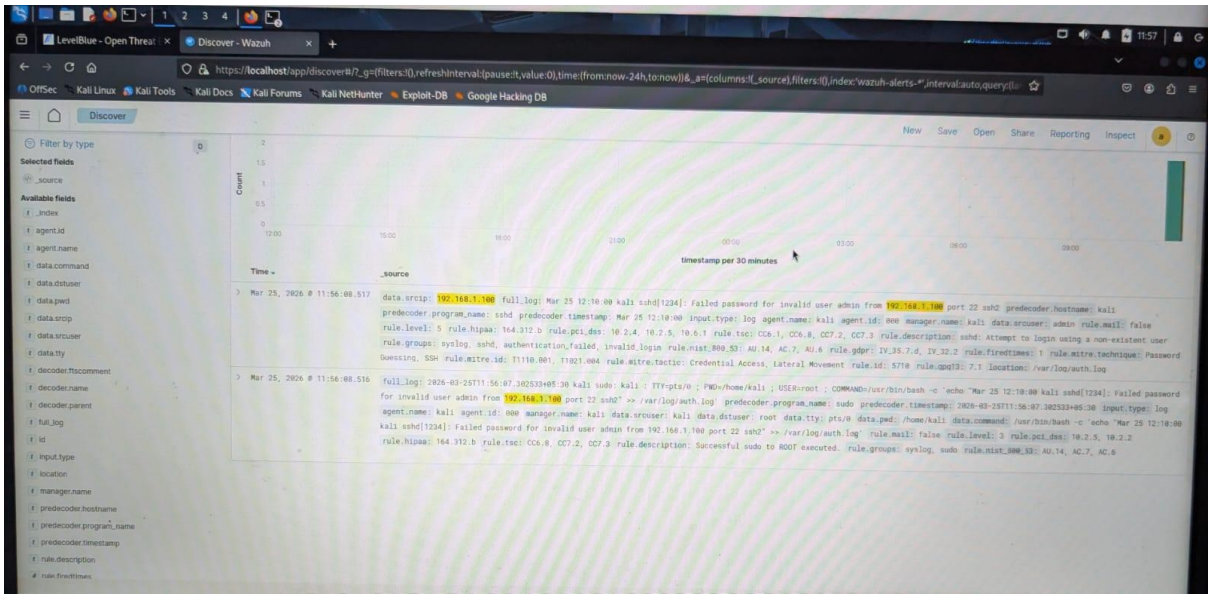
This practical exercise demonstrated how threat intelligence feeds can be integrated into a SIEM platform to enhance detection and investigation capabilities. By integrating AlienVault OTX with Wazuh, it became possible to correlate detected events with external threat intelligence indicators. The experiment showed how SOC analysts can generate alerts, enrich them with contextual intelligence, and perform threat hunting queries to identify suspicious activities within monitored systems. This integration significantly improves the effectiveness of security monitoring and incident investigation processes.


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
New Dashboard Preview
https://otx.alienvault.com/preview
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Open Threat Exchange
Dashboard
Browse
Scan Endpoints
Create Pulse
Submit Sample
API Integration
Search OTX
Visualization of Malware Clusters
Created during: last 24 hours
Select Malware Family Above for More Details
Subscribed Pulses
CanisterWorm Gets Teeth: TeamPCP's Kubernetes Wiper Targets Iran
Supply chain compromise spreads from Trivy to Checkmarx GitHub ...
KICS GitHub Action Compromised: TeamPCP Supply Chain Attack
Tracing a Multi-Vector Malware Campaign: From VBS to Open Infrast...
Pro-Iranian Nasir Security is Targeting The Energy Sector in the Mid...
GhostClaw expands beyond npm: GitHub repositories and AI workflo...
Libyan Oil Refinery Among Targets in Long-running Likely Espionage ...
Widespread GitHub Actions Tag Compromise Exposes CI/CD Secrets
CVE-2026-33017: How attackers compromised Langflow AI pipeline...
Top Community Contributors
Latest Blogs from LevelBlue
LevelBlue/Labs
See more blogs from LevelBlue
© COPYRIGHT 2026 LEVELBLUE, INC. | LEGAL | STATUS

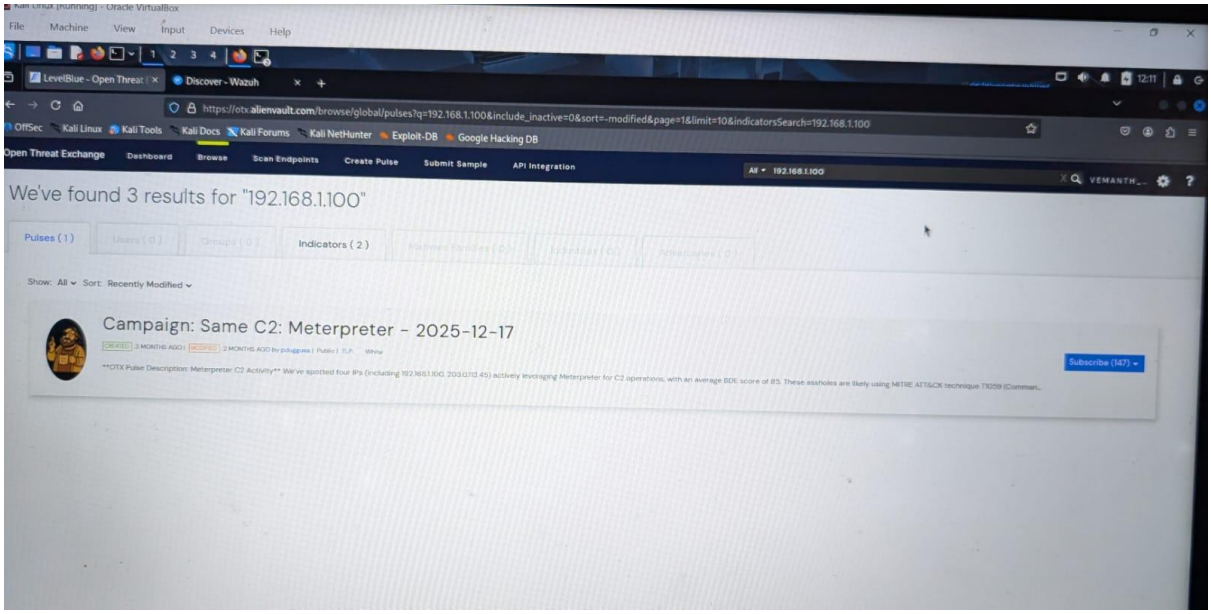

root@kali: /var/ossec
Session Actions Edit View Help
GNU nano 8.6 /var/ossec/etc/ossec.conf *
</localfile>
<localfile>
<log_format>apache</log_format>
<location>/var/log/apache2/error.log</location>
</localfile>
<localfile>
<log_format>apache</log_format>
<location>/var/log/apache2/access.log</location>
</localfile>
<localfile>
<log_format>syslog</log_format>
<location>/var/ossec/logs/active-responses.log</location>
</localfile>
<localfile>
<log_format>syslog</log_format>
<location>/var/log/auth.log</location>
</localfile>
<localfile>
<log_format>syslog</log_format>
<location>/var/log/syslog</location>
</localfile>
<localfile>
<log_format>syslog</log_format>
<location>/var/log/dpkg.log</location>
</localfile>
<localfile>
<log_format>syslog</log_format>
<location>/var/log/kern.log</location>
</localfile>
</ossec_config>
<integration>
<name>alienvault</name>
<api_key>283821956d8a9182fdc29a66c8fce3bcebde4869143e6d80f318b103bacdf733</api_key>
<group>syslog</group>
<alert_format>json</alert_format>
</integration>
^G Help ^X Exit
^O Write Out ^R Read File
^W Where Is ^\ Replace
^K Cut ^U Paste
^T Execute ^J Justify
^C Location ^/ Go To Line
29°C Sunny
Search

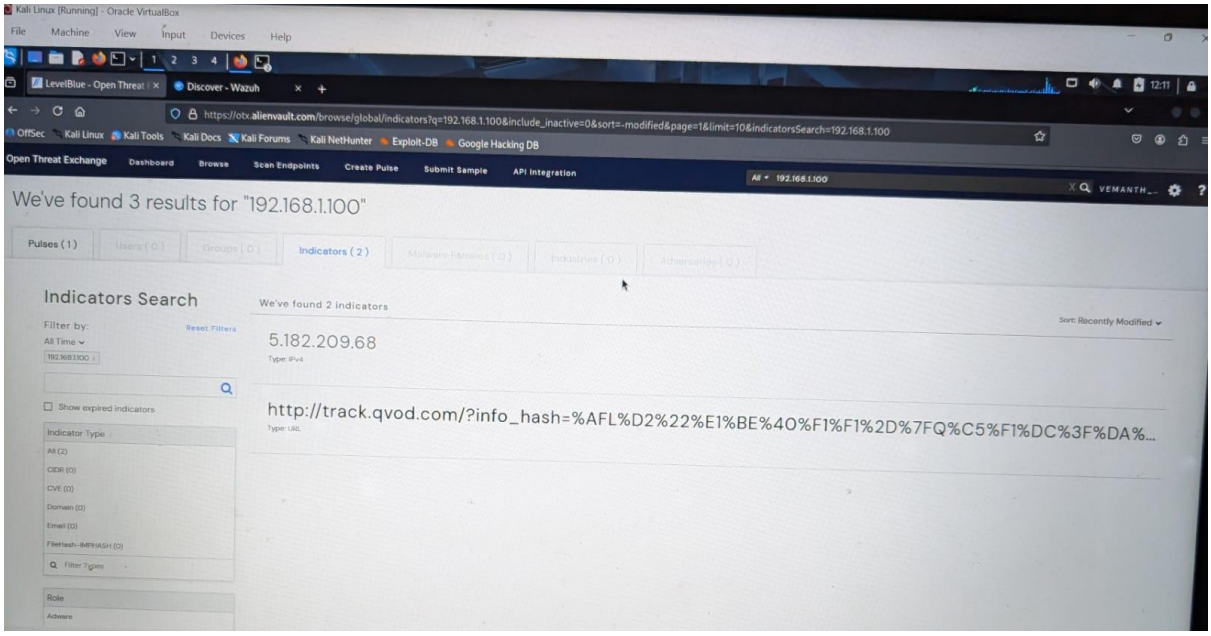
```
ress this warning.

┌──(root㉿kali)-[/var/ossec/integrations]
└─# sudo nano /var/ossec/integrations/custom-otx.py

┌──(root㉿kali)-[/var/ossec/integrations]
└─# sudo chmod +x /var/osssec/intergrations/custom-otx.py
chmod: cannot access '/var/osssec/intergrations/custom-otx.py': No su

┌──(root㉿kali)-[/var/ossec/integrations]
└─# ls /var/ossec/integrations
custom-otx.py  maltiverse.py  pagerduty.py  shuffle.py  slack.py    vi
maltiverse     pagerduty      shuffle       slack       virustotal

┌──(root㉿kali)-[/var/ossec/integrations]
└─# sudo chmod +x /var/osssec/integrations/custom-otx.py
chmod: cannot access '/var/osssec/integrations/custom-otx.py': No such

┌──(root㉿kali)-[/var/ossec/integrations]
└─# sudo chmod +x /var/ossec/integrations/custom-otx.py

┌──(root㉿kali)-[/var/ossec/integrations]
└─# ls -l /var/ossec/integrations
total 80
-rwxr-xr-x 1 root root    394 Mar 25 11:15 custom-otx.py
-rwxr-x--- 1 root wazuh  1045 May 29  2024 maltiverse
-rwxr-x--- 1 root wazuh 17358 May 29  2024 maltiverse.py
-rwxr-x--- 1 root wazuh  1045 May 29  2024 pagerduty
-rwxr-x--- 1 root wazuh  7090 May 29  2024 pagerduty.py
-rwxr-x--- 1 root wazuh  1045 May 29  2024 shuffle
-rwxr-x--- 1 root wazuh  7686 May 29  2024 shuffle.py
-rwxr-x--- 1 root wazuh  1045 May 29  2024 slack
-rwxr-x--- 1 root wazuh  7289 May 29  2024 slack.py
-rwxr-x--- 1 root wazuh  1045 May 29  2024 virustotal
-rwxr-x--- 1 root wazuh  9785 May 29  2024 virustotal.py

┌──(root㉿kali)-[/var/ossec/integrations]
└─#
```

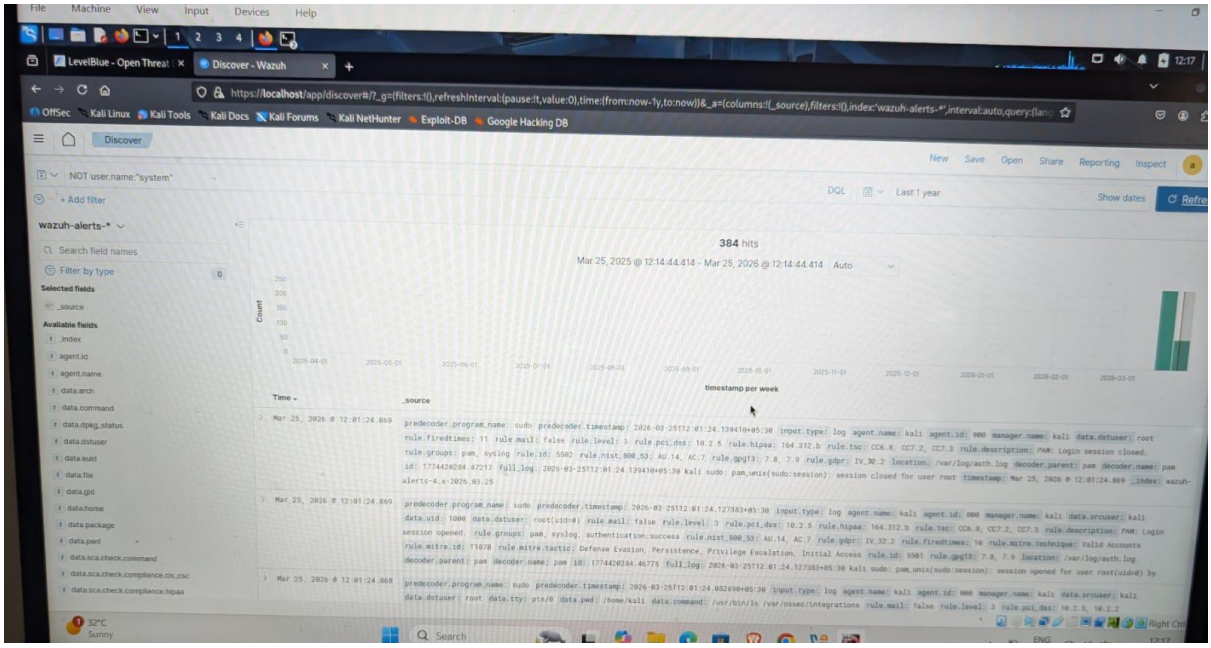
32°C
Sunny



```
^C
┌──(kali㉿kali)-[~]
└─$ sudo bash -c 'echo "Mar 25 12:10:00 kali sshd[1234]: Failed password for invalid user admin from 192.168.1.100
port 22 ssh2" >> /var/log/auth.log'

┌──(kali㉿kali)-[~]
└─$ sudo ls /var/ossec/integrations
custom-otx.py  maltiverse.py  pagerduty.py  shuffle.py  slack.py    virustotal.py
maltiverse     pagerduty      shuffle       slack       virustotal

┌──(kali㉿kali)-[~]
└─$
```






File Machine View Input Devices Help
LevelBlue - Open Threat
Discover - Wazuh
https://localhost/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-1y,to:now))&_a=(columns:!(_source),filters:!(),index:'wazuh-alerts-*',interval:auto,query:(lang
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Discover
New Save Open Share Reporting Inspect
NOT user.name:"system"
+ Add filter
DQL Last 1 year Show dates Refresh
wazuh-alerts-*
Search field names
Filter by type 0
Selected fields
_source
Available fields
_index
agent.id
agent.name
data.arch
data.command
data.dpkg_status
data.dstuser
data.euid
data.file
data.gid
data.home
data.package
data.pwd
data.sca.check.command
data.sca.check.compliance.cis_csc
data.sca.check.compliance.hipaa
384 hits
Mar 25, 2025 @ 12:14:44.414 - Mar 25, 2026 @ 12:14:44.414 Auto
Count
timestamp per week
Time
_source
Mar 25, 2026 @ 12:01:24.869
predecoder.program_name: sudo predecoder.timestamp: 2026-03-25T12:01:24.139410+05:30 input.type: log agent.name: kali agent.id: 000 manager.name: kali data.dstuser: root rule.firedtimes: 11 rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: Login session closed. rule.groups: pam, syslog rule.id: 5502 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: /var/log/auth.log decoder.parent: pam decoder.name: pam id: 1774420284.47217 full_log: 2026-03-25T12:01:24.139410+05:30 kali sudo: pam_unix(sudo:session): session closed for user root timestamp: Mar 25, 2026 @ 12:01:24.869 _index: wazuh-alerts-4.x-2026.03.25
Mar 25, 2026 @ 12:01:24.869
predecoder.program_name: sudo predecoder.timestamp: 2026-03-25T12:01:24.127383+05:30 input.type: log agent.name: kali agent.id: 000 manager.name: kali data.srcuser: kali data.uid: 1000 data.dstuser: root(uid=0) rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: Login session opened. rule.groups: pam, syslog, authentication_success rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 10 rule.mitre.technique: Valid Accounts rule.mitre.id: T1078 rule.mitre.tactic: Defense Evasion, Persistence, Privilege Escalation, Initial Access rule.id: 5501 rule.gpg13: 7.8, 7.9 location: /var/log/auth.log decoder.parent: pam decoder.name: pam id: 1774420284.46775 full_log: 2026-03-25T12:01:24.127383+05:30 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by
Mar 25, 2026 @ 12:01:24.868
predecoder.program_name: sudo predecoder.timestamp: 2026-03-25T12:01:24.052690+05:30 input.type: log agent.name: kali agent.id: 000 manager.name: kali data.srcuser: kali data.dstuser: root data.tty: pts/0 data.pwd: /home/kali data.command: /usr/bin/ls /var/ossec/integrations rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5, 10.2.2
32°C Sunny
Search
ENG
Right Ctrl